Language in different modalities: Sign and Braille

Note

Sign language and Braille are really different cases.

Sign languages, as we'll see, are natural languages that have their own linguistic structure.

Braille is a system that allows blind individuals to *read* a spoken language they share with sighted individuals.

I've lumped them together because they reveal interesting things about the plasticity of sensory systems, and how the brain gets organized differently when language input comes from different modalities.

Sign Language

Sign languages have some obvious differences from spoken languages:

- modality

- articulators

- use of space

- iconicity (though this is easy to exaggerate when you look at individual words with their glosses)



Emmorey et al. (2004, Brain & Language)

Take a look at sign language in action.



Note:

- eye contact
- coordination of hands, face, upper body
- fluidity (can you segment words?)
- speed

https://www.youtube.com/watch?v=8J_UfLH8isU

Phonology of sign language.

As with speech, there are featural descriptions of sign.

Signs can differ contrastively in:

- hand configuration

- gesture location

- movement



“please”



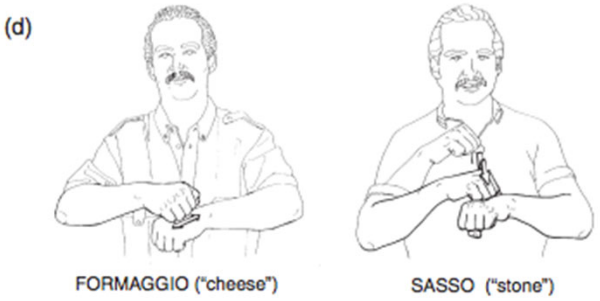
“sorry”



“onion”



“apple”



(Emmorey et al., 2003, Language & Cognitive Processes)

Research question: Is there categorical perception for these dimensions?

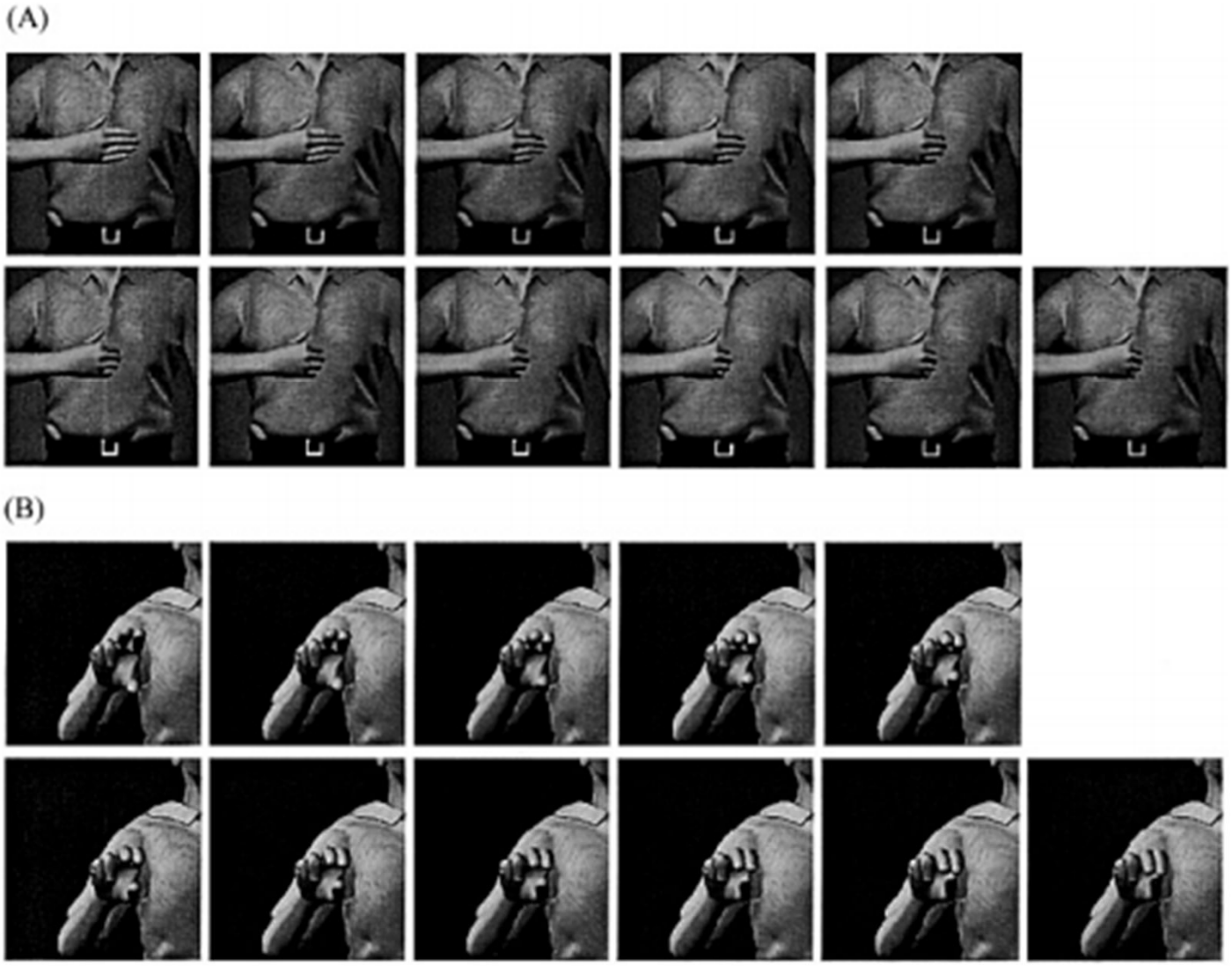
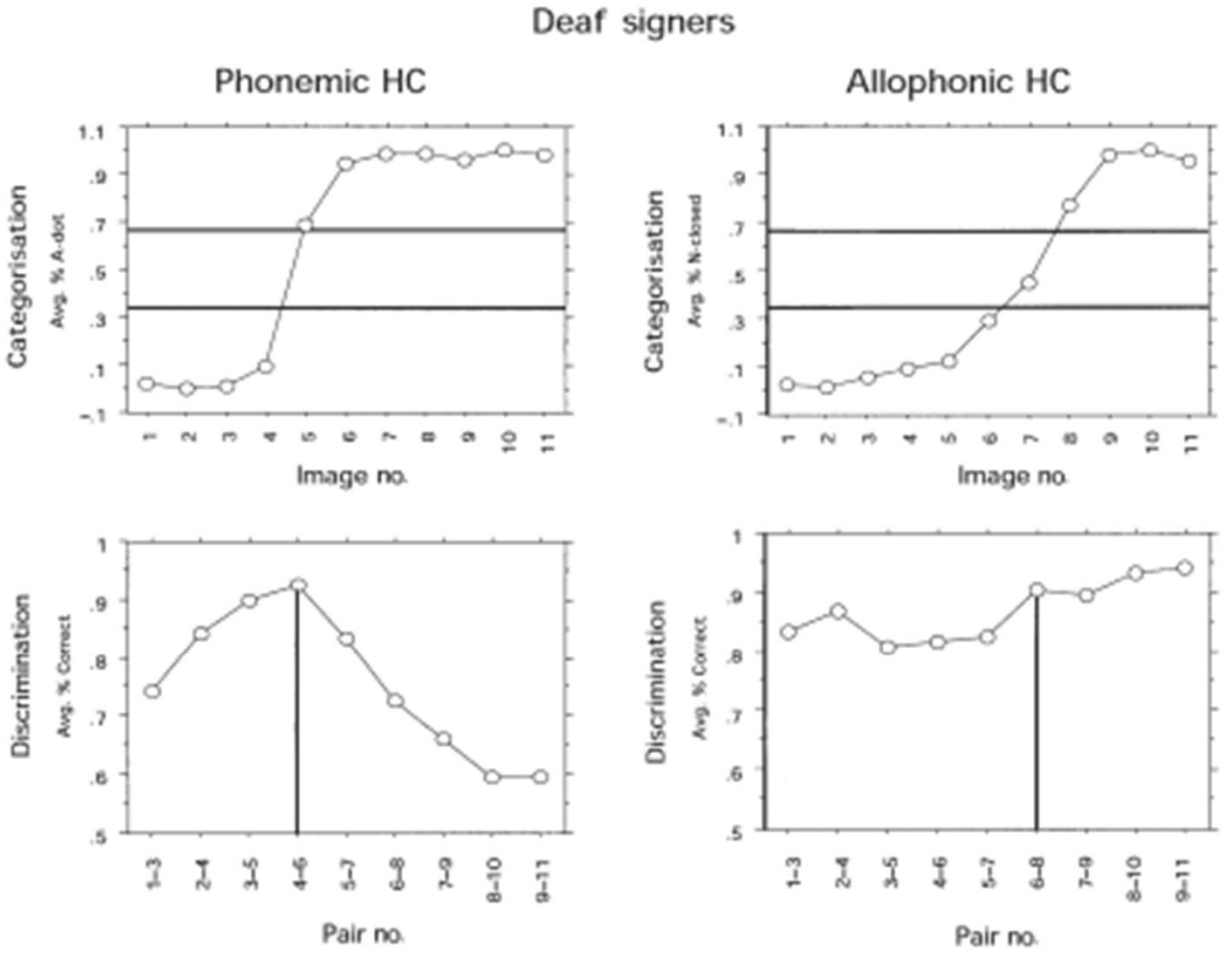


Figure 1. Illustration of stimulus continua varying in (A) phonemic hand configuration (from the sign PLEASE to the sign SORRY) and (B) allophonic hand configuration (from 'open-N' to 'closed N' in the sign SAY-NO-TO). The continua were created with Poser software from MetaCreations.



These curves should look familiar.

There is a fairly sharp categorical boundary between stimuli 4 and 6 for the phonemic boundary.

This corresponds to the peak in the discrimination function.

For the allophonic condition, the identification function is much shallower – i.e., many more ambiguous stimuli – and there is no peak in the discrimination function.

Use of spatial information in grammar.



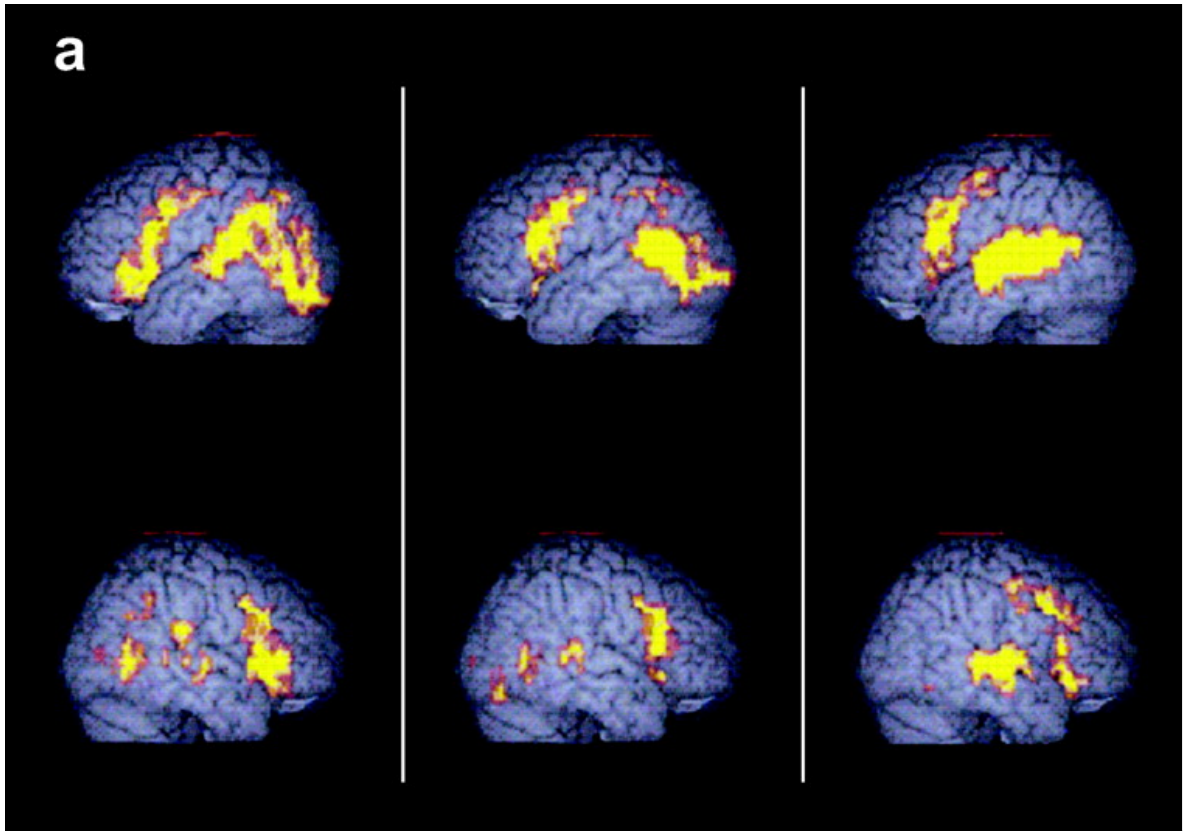
FIG. 1. An illustration of a simple spatial description in ASL, using classifier constructions. An English translation would be "The bike is near the house." Whole-entity CL refers to the type of classifier handshape morpheme and +loc refers to the position movement morpheme (a short downward movement) that means to be located.

Sign language uses space to indicate relationships between words in a sentence.

This is a very transparent example, where the relative position of two objects is given by where the classifier sign is made.

Consider that understanding sentences of any complexity in sign language requires keeping track of *where* signs were made.

Brain activity during sign language comprehension has both important similarities and differences with spoken language.



Deaf signers

Hearing signers

Audio-visual
Speech (hearing participants)

Similar:

- left lateralization
- classical language areas

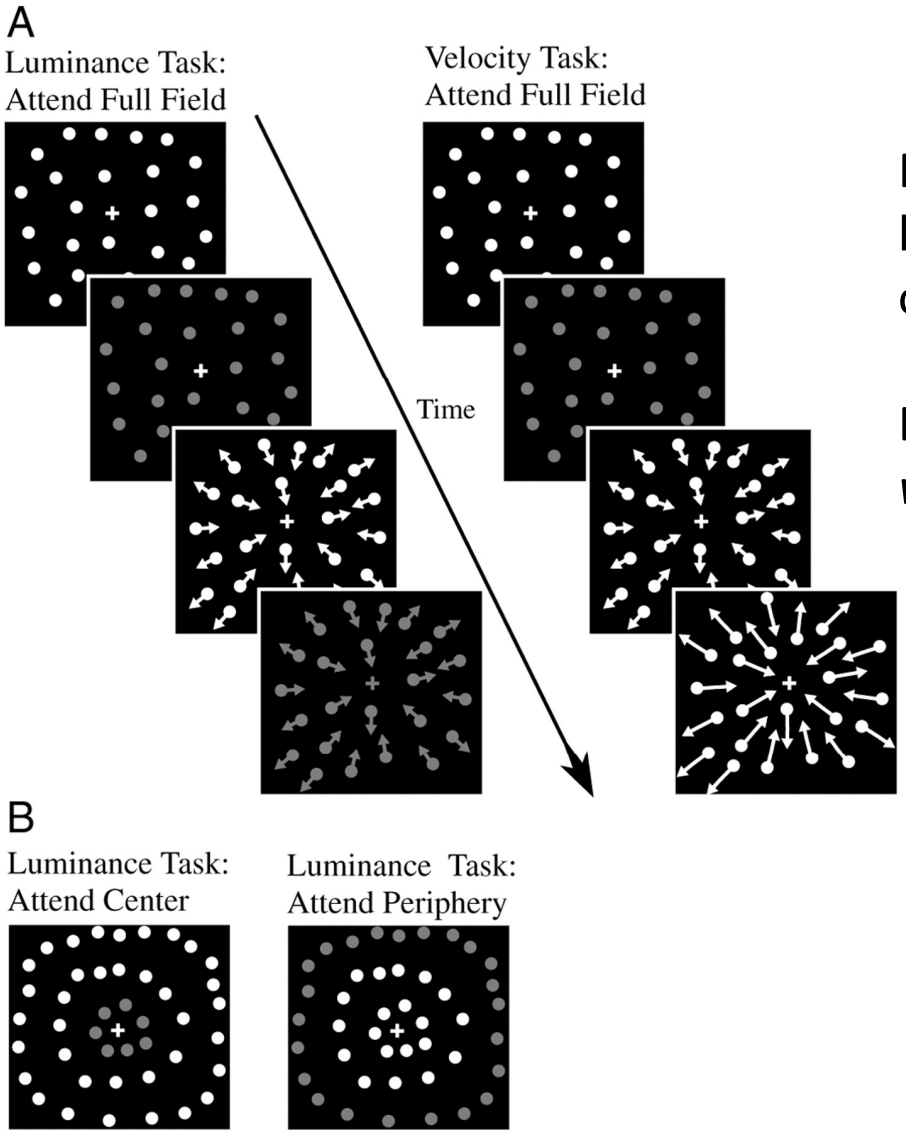
(Broca's / inferior frontal; Wernicke's / posterior superior temporal)

Different:

- visual areas
- superior parietal / classic spatial areas

Campbell, MacSweeney, & Waters (2008)

Research question:
Does deafness and/or experience with sign impact spatial attention?

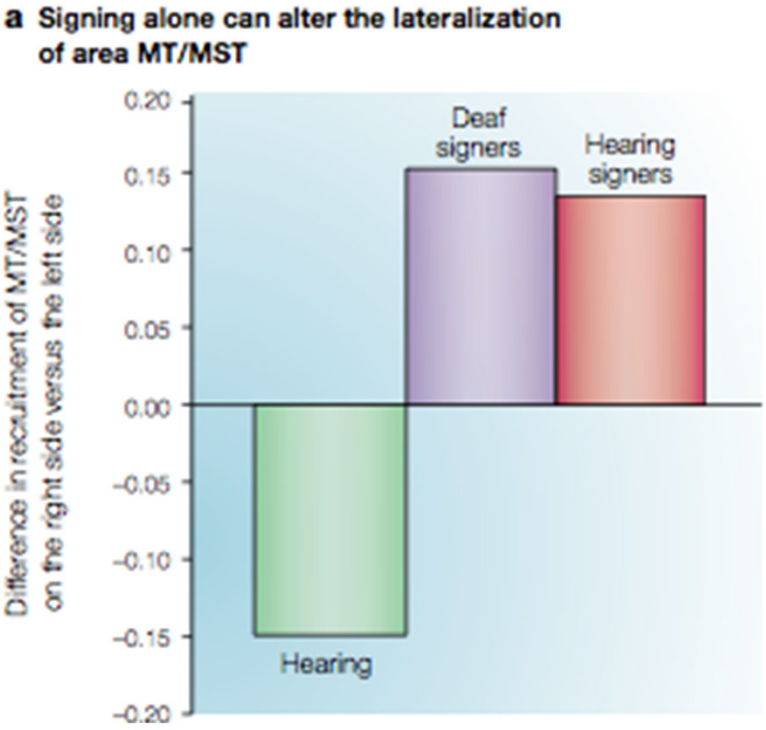


Participants monitored for brief changes in either luminance or velocity of "dots" either in the center of the screen, or at the periphery.

Deaf participants show an advantage in this task *when attention is directed to the periphery.*

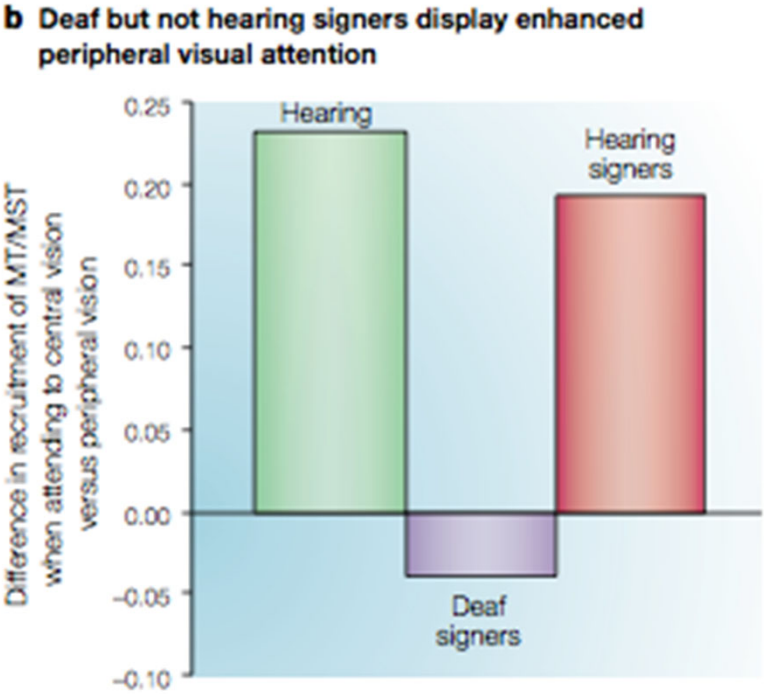
	Hearing controls	Deaf signers	Hearing signers
Number	11	9	5
Mean age	23	23	31 (22–42)
Handedness	Right	Right	Right
Audition	No loss	>90 dB loss binaurally	No loss
English skills	Native speaker	Learned at school age	Native speakers
ASL skills	None	Native signers	Native signers
Behavioral performance[1–a]			
Location of attention			
Full-field	91 (6)	78 (10)	80 (13)
Attend-center	86 (7)	78 (10)	90 (10)
Attend-periphery	82 (8)	94 (5)	80 (13)
Attentional task			
Luminance	91 (5)	89 (6)	87 (9)
Velocity	82 (7)	78 (8)	80 (11)

Bavelier D et al. J. Neurosci. 2001;21:8931-8942



Learning sign language changes the laterality of areas involved in motion perception from right to left.

(Perhaps because motion perception needs to "talk to" left-lateralized language networks in people who learn sign.)



Enhancements of peripheral visual attention, however, are only observed in deaf individuals.

(Perhaps because hearing individuals are more dependent on auditory cues for spatial attention when things are at the far periphery.)

Dissociating visual/spatial/motor processing of hand movements from language processing.



Fig. 2. Examples of stimuli. (Top row) Still images from a Tic Tac sentence. (Bottom row) Still images from a BSL sentence—English translation: 'This building is being renovated.'

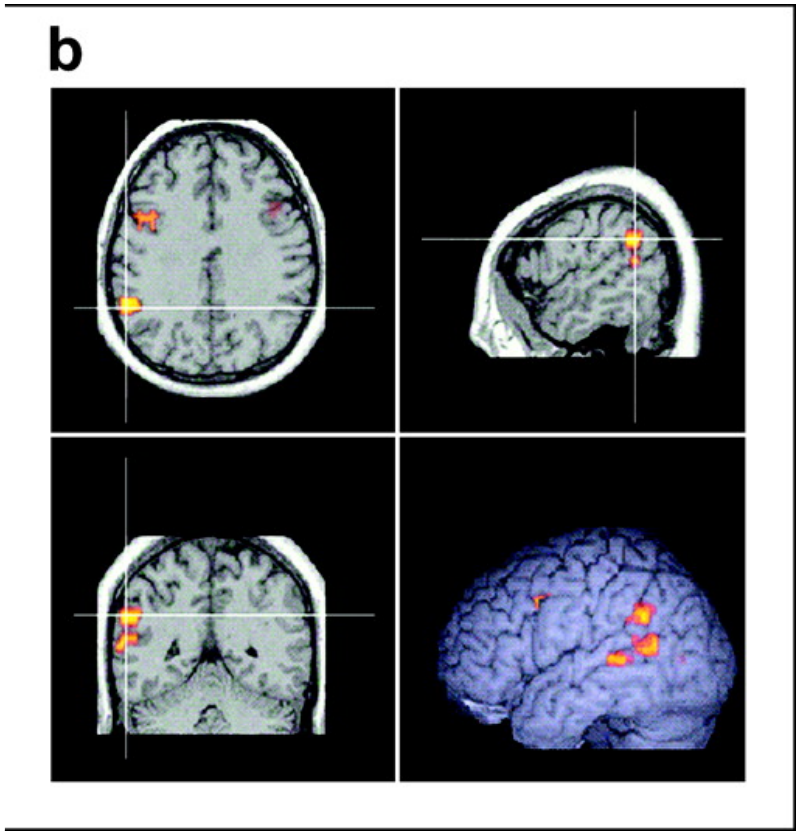
Direct comparison of sign comprehension with "tic-tac," a sophisticated (and outmoded) gestural system used to communicate about odds at horse races in the UK.

https://www.youtube.com/watch?v=YsZu_wZa-j8

Stronger activity was observed around Wernicke's area for sign language vs. tic-tac.

(The frontal activation is superior to Broca's area; you can make up a "hand-wavy" story about why tic-tac might activate Broca's area.)

(See what I did there? Hand-wavy? Ha ha.)



Campbell, MacSweeney, & Waters (2008)

So, sign language…

Sign language has phonological structure, and native speakers show categorical perception for some contrastive features.

Learning a signed language results in some brain reorganization, particularly the lateralization of motion processing.

Although attention to the visual periphery is critical for both hearing and deaf signers, only deaf signers show enhanced processing of visual attention in the periphery.

Compared with non-language stimuli that also involve organized, sequenced hand movements, signed sentences activate posterior STG (Wernicke's area).

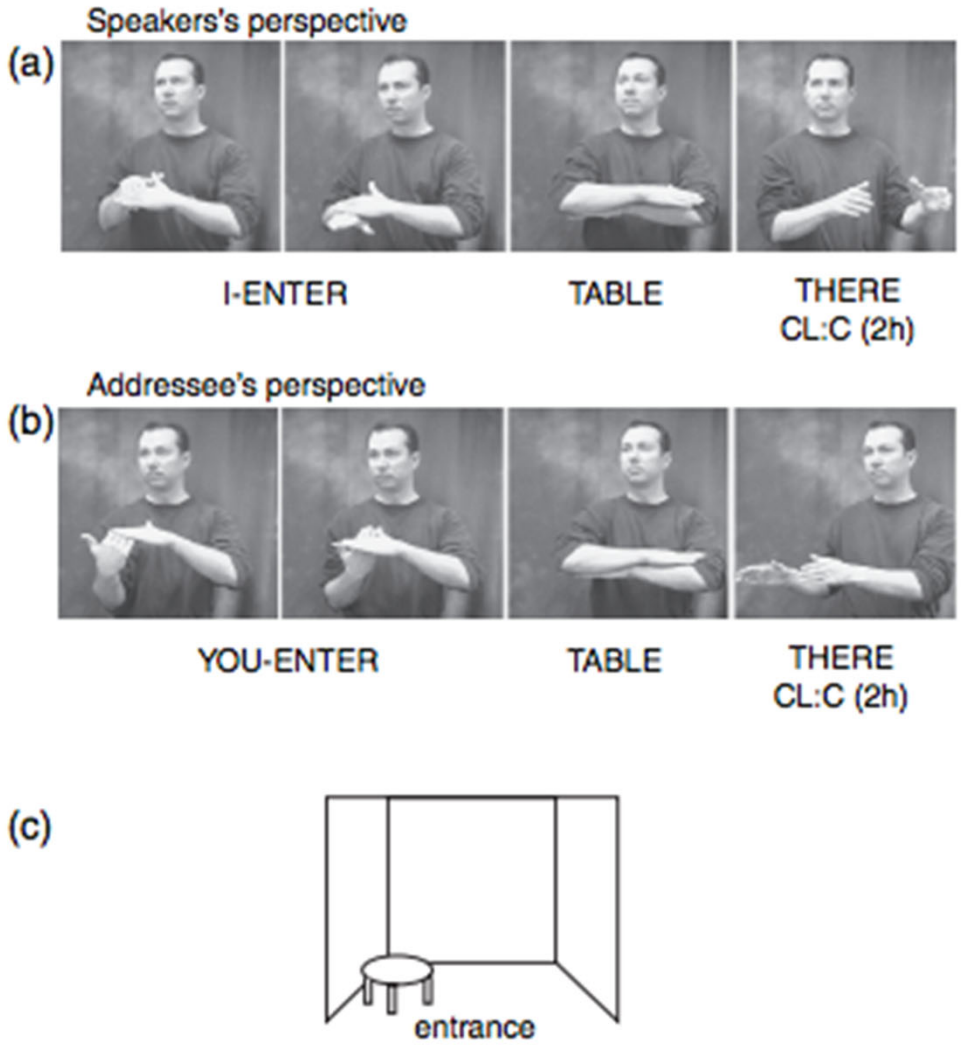
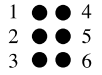


Figure 43.4 Illustration of ASL descriptions of the location of a table within a room, described from (A) the speaker's perspective or (B) the addressee's perspective. Signers exhibit better comprehension for room descriptions presented from the speaker's perspective, despite the mental transformation that this description entails. Reprinted from Emmorey (2002), with permission.

Tactile reading: Braille

Braille Alphabet

The six dots of the braille cell are arranged and numbered:

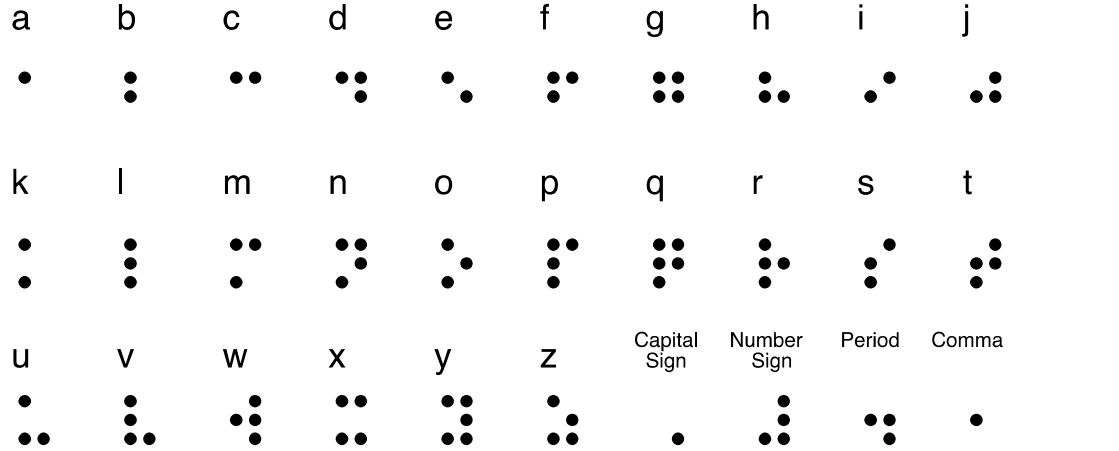


The capital sign, dot 6, placed before a letter makes a capital letter.



The number sign, dots 3, 4, 5, 6, placed before the characters a through j, makes the numbers 1 through 0. For example: a preceded by the number sign is 1, b is 2, etc.

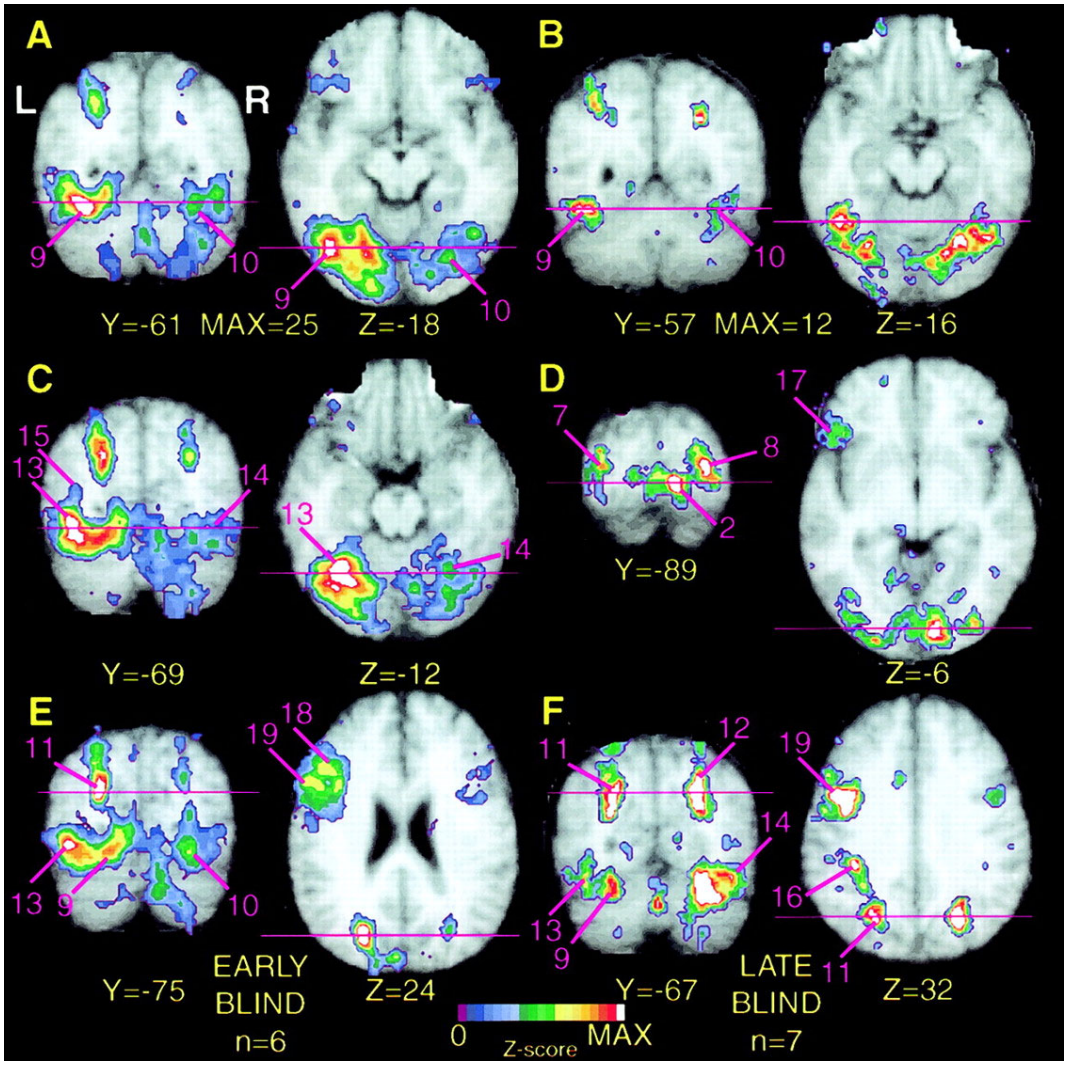




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Braille reading is in many ways equivalent to reading print, BUT:

- symbols are optimized for tactile discrimination (reading Braille by sight is really hard)
- fingers take in letters one at a time



Reading Braille activates visual cortex in blind individuals.

The images on the left show three early blind participants, with late blind participants on the right.

The earlier sight is lost, the more extensive visual activity is to tactile stimuli.

We haven't talked about the "visual word form area," but it's strongly activated in early blind participants, more so than late blind.

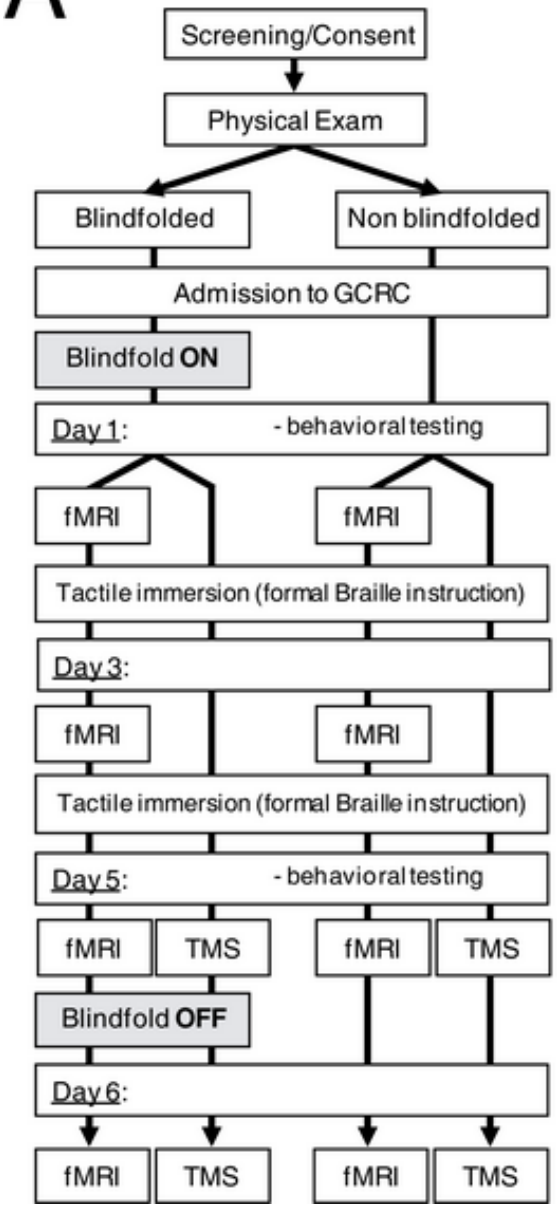
A question about competition

Remember that we worried about whether competition could be preventing hearing individuals from developing awesome peripheral visual attention?

Merabet et al. asked sighted individuals to wear blindfolds for *five days,* and trained them to read Braille.

Relative to a control group, the blindfolded group learned more quickly.

A



B





Merabet et al. (2008, PLoS ONE)

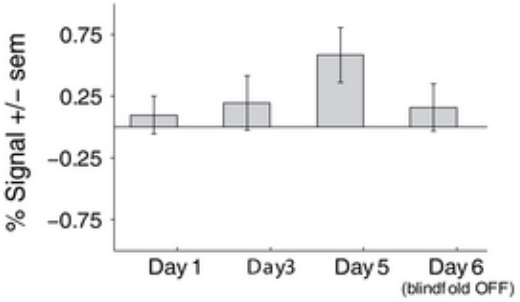
A



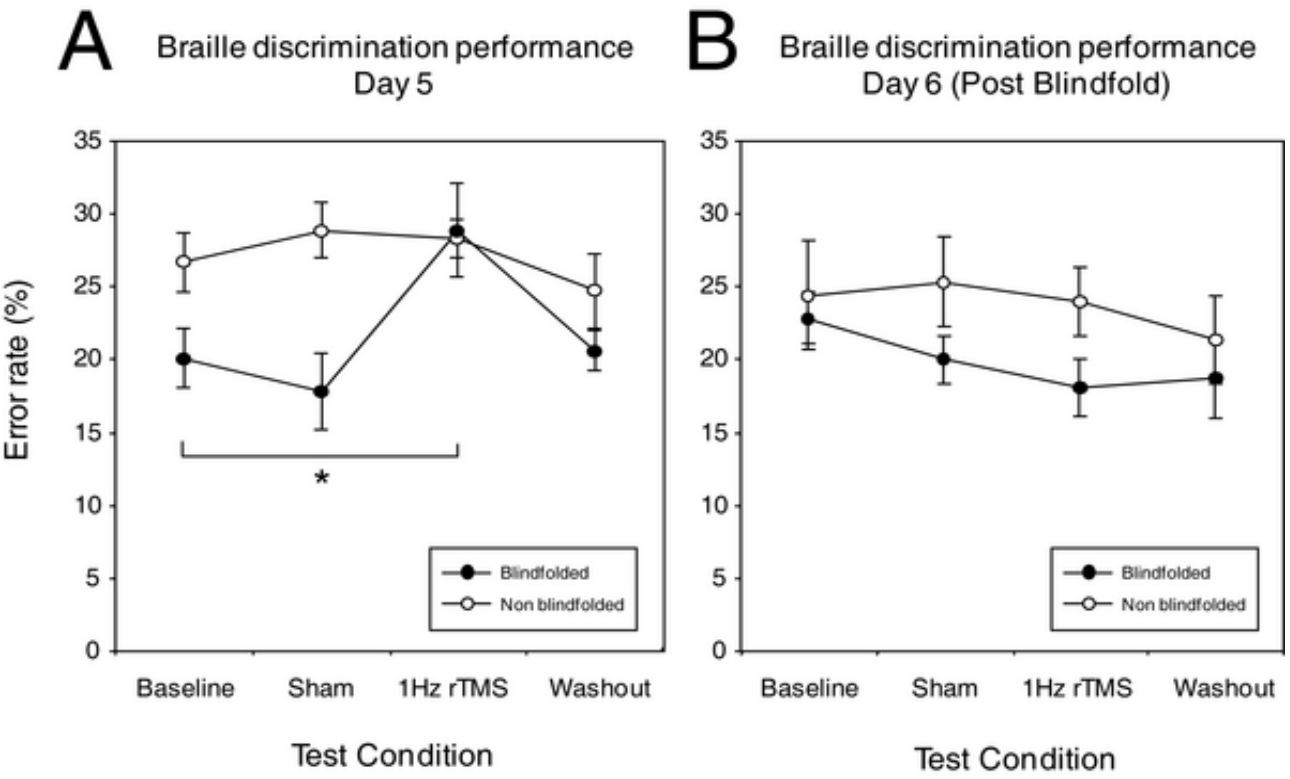
Further, over the course of the experiment, sighted, blindfolded participants started to show some activity in primary *visual* cortex when reading Braille.

When the blindfold was removed on Day 6, the effect was completely reversed.

B



Merabet LB, Hamilton R, Schlaug G, Swisher JD, et al. (2008) Rapid and Reversible Recruitment of Early Visual Cortex for Touch. PLoS ONE 3(8): e3046. doi:10.1371/journal.pone.0003046
http://www.plosone.org/article/info:doi/10.1371/journal.pone.0003046



When transcranial magnetic stimulation (TMS) was used to disrupt activity in visual cortex, it disrupted reading *in the blindfolded group only* (solid symbols).

This is evidence that the blindfolded participants are somehow *using visual cortex* for tactile processing. (Maybe?)



One way to think about this is that visual cortex is getting tactile input all the time, even in sighted, unblindfolded people, but that this input is too weak to *compete* with the much stronger, more consistent visual input.

Even short-term removal of visual input can “unmask” tactile responses in visual cortex.

Cross-modal plasticity and language

An important point is the *resilience* of language circuitry despite differences in modality.

- Classical language areas are activated by sign language, although the input is dramatically different in its surface form.
- Braille reading seems to engage networks that are important for processing text.

We can think of cross-modal plasticity as demonstrating that brain organization is shaped over the course of development by *competition.*

For example, when auditory input is unavailable, putative auditory areas are used to process sign, and visual attention is enhanced at the periphery.

Recall that the competing cognitive/perceptual processes shaping language processing were also important when we considered the *Competition Model* as relevant to *second language learning.*